

The Learnability of Preantepenultimate Stress: Insights from Indo-Aryan Languages on the Typology and Diachrony of Stress Systems

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Guiding Questions

1. Do there exist stress systems that permit *metrically assigned* primary stress within a “window” of four syllables?

(1) **Def.: STRESS WINDOW**

A distance in syllables (σ) from the left or right edge of a prosodic word (ω) within which primary stress is permitted to occur.

- In a right-edge oriented four-syllable window, primary stress is allowed on the *preantepenultimate* syllable, but not farther to the left.

Yes $[\sigma' \sigma \sigma \sigma]$

No $*[\sigma' \sigma \sigma \sigma]$

2. Given a positive answer to question 1., how is the *comparative rarity* of such systems to be explained?

- This talk will consider role of LEARNABILITY in LANGUAGE CHANGE for understanding the typology of word-prominence systems (metrical stress more specifically).
- The **Indo-Aryan** languages — spanning from Vedic Sanskrit to the Modern languages — contribute important data to this typology, and how change in this domain may occur.

Focus Language: Sanskrit

- ▶ **Sanskrit** is the one of the earliest-attested Indo-European languages, a classical language of the Indian subcontinent, and an ancestor of modern Indo-Aryan languages (Hindi, Gujarātī, Marāṭhī, etc.). Two varieties:
 - ▶ **Vedic Sanskrit** (c. 1500–500 BCE): language of the oldest Sanskrit texts, a living language, dialectal differences discernible (Brereton and Jamison 2020).
 - ▶ **Classical Sanskrit** (c. 500 BCE –): grammatically fixed form, written and spoken alongside vernacular languages descended from Vedic (Prākṛits).
- ▶ Good reasons to believe that some varieties of post-Vedic Sanskrit — and by tradition, spoken Classical Sanskrit in some areas — exhibited a stress pattern predictable based on syllable count and syllable weight.

(2) Stress in Classical Sanskrit

- ja.(‘gau) ‘go:PERF.1SG’
- (,ka.ra).(‘ṇē).(nā) ‘doing:INST.SG’
- (,u.pa).(‘ga.ma).(tam) ‘approach:AOR.2DU’
- (‘du.hi).ta.(ram) ‘daughter:ACC.SG’

⇒ How exactly does this stress system function, is it genuinely learnable, and why are comparable systems so uncommon?

Sanskrit and the History of Phonology

- ☞ The study of Sanskrit has yielded a lot of insights in the development of aspects of phonological theory!
- ▶ **Natural classes for place/manner of articulation** (*prāṭiśākyas*; Whitney 1868 [1973], Shastri 1922; cf. Allen 1953)
 - ▶ **sandhi**: phonological processes whose application must be defined by reference to prosodic domains (Allen 1962 [1972]; cf. Selkirk 1980)
 - ▶ **Context-sensitive rewrite rules**: $A \rightarrow B / C_D$ (Pāṇinian tradition; cf. Kiparsky 1993)
 - ▶ **Level-ordering in Lexical Phonology** (“primary” vs. “secondary” derivational affixes in Whitney 1889 [1960] and Macdonell 1910; cf. Kiparsky 1982:5; 73–82)
 - ▶ **Dominance effects in lexical stress assignment** Kiparsky and Halle 1977, Kiparsky 2010)
 - ▶ **Today’s Major Point**: Sanskrit — together with descendent Indo-Aryan languages — makes an important contribution to our understanding of metrical stress systems and their learnability.



Basics of Stress Typology

- ▶ How WORD-PROMINENCE SYSTEMS (“stress”, “accent”, etc.) function is among the most thoroughly studied phonological questions:
 - ▶ Typologically (e.g., Goedemans and van der Hulst 2009, Goedemans et al. 2014)
 - ▶ Theoretically (e.g., Hayes 1995, Halle and Idsardi 1995)
 - ▶ Computationally (e.g., Tesar and Smolensky 2000, Heinz 2009)
- ▶ Three general categories of *morphologically insensitive* stress grammars:
 1. Bounded quantity-insensitive patterns (= only syllable count and distance from left/right edge of ω matters), e.g., Icelandic (Árnason 1985).
 2. Bounded quantity-sensitive patterns (= some further property [syllable weight, tone, lexical accent] can preferentially attract stress), e.g., Classical Latin: primary stress on heavy penult, else antepenult (Mester 1994).
 3. Unbounded quantity-sensitive patterns, e.g., Cupeño: primary stress on any syllable, determined by lexical accents (Uto-Aztecan; Yates 2017:Ch. 3).
- ▶ Languages belonging to categories 1. or 2. have primary stress within a WINDOW; languages belonging to category 3. lack a stress window.

Simple Example: Initial Primary Stress in Pintupi (1)

- **Pintupi** (Pama-Nyungan; Wati; Australia): bounded, quantity-insensitive. Place primary stress on the initial syllable of the ω ; place secondary stress on every second syllable thereafter (unless it is the final syllable).

(3) Left-Oriented SYLLABIC TROCHEES in Pintupi (Hansen and Hansen 1969; cf. also Hayes 1995:62–4, Apoussidou 2007:Ch. 5)

- a. [ˈpu.liŋ.ka.la.t̪u] ‘we (sat) on the hill’
- b. [ˈt̪a.mu.liŋ.pa.t̪uŋ.ku] ‘our relation’

⇒ How do we go about generating exactly this pattern of stress?

Simple Example: Initial Primary Stress in Pintupi (2)

- ▶ Patterns of metrical stress have been a poster-child for **phonological analysis through constraint interaction** (Optimality Theory and related models) since its inception (Prince and Smolensky 1993 [2004]:Ch. 4).
- ▶ Five important properties of Pintupi stress (cf. Kager 1999:Ch. 4 for constraints):
 - ▶ Feet have prominence on the **left** = trochaic feet (TROCHEE $\gg$ IAMB):
*[(pu.'lɪŋ).(ka.'la).tʃu].
 - ▶ Feet are built as close to the **left** edge of the word as possible (ALL-FEET-LEFT $\gg$ ALL-FEET-RIGHT): *[pu.('lɪŋ.ka).('la.tʃu)]
 - ▶ **Leftmost** foot in the word is the **HEAD** foot of the prosodic word (LEFTMOST $\gg$ RIGHTMOST): *[('pu.lɪŋ).('ka.la).tʃu]
 - ▶ Feet (F) are strictly **binary** = (σ σ) (FOOTBINARITY): *[('pu.lɪŋ).('ka.la).('tʃu)]
 - ▶ As many syllables as possible should be included (“parsed”) into foot constituents (PARSE- σ $\gg$ ALL-FEET-LEFT): *[('pu.lɪŋ).ka.la.tʃu]
- ▶ **Total Constraint Ranking:** {TROCHEE, LEFTMOST, FOOTBINARITY} $\gg$ PARSE- σ $\gg$ {ALL-FEET-LEFT, RIGHTMOST, IAMB} $\gg$ ALL-FEET-RIGHT

Simple Example: Initial Primary Stress in Pintupi (3)

(3) /pu.liŋkalat̪u/ → [(ˈpu.liŋ).(ˈka.la).t̪u] ‘we (sat) on the hill’

/pu.liŋkalat̪u/		TROCHEE	LEFTMOST	FOOTBINARITY	PARSE-σ	ALL-FEET-LEFT	IAMB	RIGHTMOST	ALL-FEET-RIGHT
a.	☞ (ˈpu.liŋ).(ˈka.la).t̪u				*	**	**	*	***
b.	pu.(ˈliŋ.ka).(ˈla.t̪u)				*	***!*	**	*	**
c.	(ˈpu.liŋ).ka.la.t̪u				**!*				***
d.	(ˈpu.liŋ).(ˈka.la).(t̪u)			*!		*****	**	*	***
e.	(ˈpu).(liŋ.ka).(ˈla.t̪u)			*!		*****	**	*	*****
f.	(pu.liŋ).(ˈka.la).t̪u		*!		*	**	**		***
g.	(pu.liŋ).(ka.ˈla).t̪u	*!*			*	**		*	***

► Candidate g. has iambic feet, not trochaic feet.

Simple Example: Initial Primary Stress in Pintupi (3)

(3) /pu.liŋkaɭa.tu/ → [(ˈpu.liŋ).(ˈka.la).tu] ‘we (sat) on the hill’

/pu.liŋkaɭa.tu/		TROCHEE	LEFTMOST	FOOTBINARITY	PARSE-σ	ALL-FEET-LEFT	IAMB	RIGHTMOST	ALL-FEET-RIGHT
a.	ⓘ (ˈpu.liŋ).(ˈka.la).tu				*	**	**	*	***
b.	pu.(ˈliŋ.ka).(ˈla.tu)				*	***!*	**	*	**
c.	(ˈpu.liŋ).ka.la.tu				***!*				***
d.	(ˈpu.liŋ).(ˈka.la).(ˈtu)			*!		*****	**	*	***
e.	(ˈpu).(liŋ.ka).(ˈla.tu)			*!		*****	**	*	*****
f.	(pu.liŋ).(ˈka.la).tu		*!		*	**	**		***
g.	(pu.liŋ).(ka.la).tu	*!*			*	**		*	***

► Candidate f. has primary stress on the rightmost foot, not the leftmost foot.

Simple Example: Initial Primary Stress in Pintupi (3)

(3) /puɭɪŋkalatɪu/ → [(^ˈpu.ɭɪŋ).(ka.la).tɪu] ‘we (sat) on the hill’

/puɭɪŋkalatɪu/		TROCHEE	LEFTMOST	FOOTBINARITY	PARSE-σ	ALL-FEET-LEFT	IAMB	RIGHTMOST	ALL-FEET-RIGHT
a.	ᵐᵒᵐᵒ (^ˈ pu.ɭɪŋ).(ka.la).tɪu				*	**	**	*	***
b.	pu.(^ˈ ɭɪŋ.ka).(la.tɪu)				*	***!*	**	*	**
c.	(^ˈ pu.ɭɪŋ).ka.la.tɪu				***!				***
d.	(^ˈ pu).(ɭɪŋ.ka).(la.tɪu)			*!		*****	**	*	*****
e.	(^ˈ pu.ɭɪŋ).(ka.la).(tɪu)			*!		*****	**	*	***
f.	(pu.ɭɪŋ).(ka.la).tɪu		*!		*	**	**		***
g.	(pu.ɭɪŋ).(ka.la).tɪu	*!*			*	**		*	***

- Candidates **d.** and **e.** each have a degenerate, monosyllabic foot (and e. is harmonically worse than d., because it incurs more violations of ALL-Ft-L).

Simple Example: Initial Primary Stress in Pintupi (3)

(3) /pu.liŋkalatju/ → [(¹pu.liŋ).(ka.la).tju] 'we (sat) on the hill'

/puḷiŋkalat̪u/		TROCHEE	LEFTMOST	FOOTBINARITY	PARSE-σ	ALL-FEET-LEFT	IAMB	RIGHTMOST	ALL-FEET-RIGHT
a.	 ('pu.liŋ).(ka.la).t̪u				*	**	**	*	***
b.	pu.('liŋ.ka).(la.t̪u)				*	***!*	**	*	**
c.	('pu.liŋ).ka.la.t̪u				***!*				***
d.	('pu.liŋ).(ka.la).(t̪u)			*!		*****	**	*	***
e.	('pu).(liŋ.ka).(la.t̪u)			*!		*****	**	*	*****
f.	(pu.liŋ).('ka.la).t̪u		*!		*	**	**		***
g.	(pu.liŋ).(ka.la).t̪u	*!*			*	**		*	***

- Candidate c. contains only a single foot, when it would be possible to build another without violating FOOTBINARITY.

Simple Example: Initial Primary Stress in Pintupi (3)

(3) /puɭɪŋkalatɕu/ → [(ˈpu.ɭɪŋ).(ˈka.la).tɕu] ‘we (sat) on the hill’

/puɭɪŋkalatɕu/	TROCHEE	LEFTMOST	FOOTBINARITY	PARSE-σ	ALL-FEET-LEFT	IAMB	RIGHTMOST	ALL-FEET-RIGHT
a.  (ˈpu.ɭɪŋ).(ka.la).tɕu				*	**	**	*	***
b. pu.(ˈɭɪŋ.ka).(ˈla.tɕu)				*	***!*	**	*	**
c. (ˈpu.ɭɪŋ).ka.la.tɕu				***!				***
d. (ˈpu.ɭɪŋ).(ka.la).(ˈtɕu)			*!		*****	**	*	***
e. (ˈpu).(ɭɪŋ.ka).(ˈla.tɕu)			*!		*****	**	*	*****
f. (ˈpu.ɭɪŋ).(ˈka.la).tɕu		*!		*	**	**		***
g. (pu.ɭɪŋ).(ka.la).tɕu	*!*			*	**		*	***

- Candidate **b.** builds its feet closer to the right edge of the word than the left edge of the word.

Simple Example: Initial Primary Stress in Pintupi (3)

(4) /pu[ɪŋkalatʃu/ → [(ˈpu.ɪŋ).(ˈka.la).tʃu] ‘we (sat) on the hill’

/pu[ɪŋkalatʃu/		TROCHEE	LEFTMOST	FOOTBINARITY	PARSE-σ	ALL-FEET-LEFT	IAMB	RIGHTMOST	ALL-FEET-RIGHT
a.	☞ (ˈpu.ɪŋ).(ˈka.la).tʃu				*	**	**	*	***
b.	pu.(ˈɪŋ.ka).(ˈla.tʃu)				*	***!*	**	*	**
c.	(ˈpu.ɪŋ).ka.la.tʃu				**!*				***
d.	(ˈpu.ɪŋ).(ˈka.la).(ˈtʃu)			*!		*****	**	*	***
e.	(ˈpu).(ˈɪŋ.ka).(ˈla.tʃu)			*!		*****	**	*	*****
f.	(ˈpu.ɪŋ).(ˈka.la).tʃu		*!		*	**	**		***
g.	(pu.ˈɪŋ).(ka.la).tʃu	*!*			*	**		*	***

- ▶ Note that this system is *simple* because several constraints (LEFTMOST, ALL-Ft-LEFT, TROCHEE) all help to unambiguously point to primary stress on the initial syllable.
- ▶ We'll return later to **quantity-sensitive** and **unbounded** stress patterns.

Uneven Distributions in Stress Typology

- **Crucially:** Not all stress patterns / windows are equally likely! (Hyman 1977)
- According to Kager's (2012: 1464) survey of stress windows of any kind, the following frequency distribution of window systems obtains:

(5) Frequency of stress window sizes in Goedemans and van der Hulst 2009

Window Size	Phonological	Lexical	Mixed	Total
Final two σ	59	20	3	82
Final three σ	22	12	4	38
Initial two σ	26	10	3	39
Initial three σ^*	1	0	0	1
Total	108	42	10	160

- Where are the (final) four-syllable windows? Should they be here?

Should Four-Syllable Windows Exist?

- ▶ Vaux (2021) voices skepticism of four-syllable windows generally, while Pater (2009:1017–21) and Kager (2012) issue (qualified) denials of four-syllable windows.
 - ▶ There do clearly seem to be some cases of four-syllable windows involving lexical stress (e.g., Romanian, Italian; cf. Roca 1999:666–83).
 - ▶ Some further languages permit preantepenultimate stress when prominence considerations block primary stress on the penultimate or antepenultimate syllable (Ashéninca, Payne 1990; Paamese, Lee 1999).
- 👉 But the real interest at present lies with *predictable metrical stress systems* in which no additional exceptional considerations apply.

Cases of Four-Syllable Metrical Stress Windows

- ▶ Possible cases of predictable metrical stress systems with four-syllable stress windows (all right-edge oriented):
 - ▶ Old Latin: evidence largely based on metrics
(Corssen 1870:892–906, Sturtevant 1919, Allen 1973:188–90, Jacobs 2003; but cf. Fortson 2008:30–3)
 - ▶ Classical Arabic and some modern varieties of Arabic: data reported often difficult to distinguish from a default-to-opposite system.
(Watson 2011:§4, Janssens 1972, Brockelmann 1907)
 - ▶ Post-Vedic (“Classical”) Sanskrit
(Sandell 2023b)

- ▶ **Empirical Claim:** Predictable four-syllable windows *do* exist, and probably constitute the upper limit for possible patterns of bounded stress.
 - ▶ Such a window is instantiated in a post-Vedic Indo-Aryan language reflected in data reported for Classical Sanskrit and reconstructable based on Gujarātī.
- ▶ **Theoretical Claim:** Four-syllable windows in predictable stress systems result mainly from two opposing edge preferences: primary stress should fall on the {left/right}-most foot ($\mathfrak{f}$) in a prosodic word (ω), but the opposite edge of a ω should ideally be aligned with a foot.
- ▶ **Typological and Historical Claim:** The rarity of four-syllable stress windows testifies to both 1) an **analytic simplicity bias** in the learning of phonological grammars and to 2) **frequency sensitivity** on the part of learners.
 1. Phonological grammars capable of generating larger windows involve more intricate constraint interactions.
 2. Accurately acquiring more intricate constraint interactions requires more data.
 3. Diachronically, an analytic simplicity bias and frequency sensitivity act as a “filter” on intricate stress patterns.

Research Methodology (1)

- **Overall Approach:** Integration of research phonological learnability, phonological typology, and phonological change, applied to Indo-Aryan languages.

1. Phonological Learnability: Do there exist provably correct procedures (in a computational sense) that are able to generate the data seen in a grammar?

Tesar and Smolensky 2000, Goldwater and Johnson 2003, Apoussidou 2007, Albright and Hayes 2011, Jarosz 2013, Tesar 2014, Jarosz 2019

2. Phonological Typology: What types of phonological systems can/cannot exist, which ones are more/less common, and why?

Many surveys of stress patterns available: Bailey 1995, Gordon 2002, Goedemans and van der Hulst 2009, Goedemans et al. 2014

3. Phonological Change: How does Grammar (+ Lexicon) $\mathcal{A}$ become Grammar (+ Lexicon) $\mathcal{B}$?

Kiparsky 1968, Kiparsky 2003, Wedel 2006, Kiparsky 2015, Wedel 2015, Hayes 2015, Kiparsky 2016, Harrington et al. 2018

Research Methodology (2)

- ▶ **MAXIMUM ENTROPY GRAMMAR:** simulation using constraint-based grammars, in which the constraints have real-valued weights, rather than strict rankings (Goldwater and Johnson 2003, Jäger 2007).
 - ▶ Can fit grammars with synchronically variable outputs $\Rightarrow$ appropriate to modeling language change (Hayes 2022).
- ▶ **ROBUST INTERPRETIVE PARSING:** detecting mismatches between learning data and outputs of a grammar to discover “hidden” phonological structure (Tesar and Smolensky 1998).
- ▶ **(ITERATED) AGENT-BASED MODELING:** learning agents generalize from data and provide data to other learning agents (Kirby and Hurford 2002).

Seed Data $\rightarrow$ Learner 1 $\rightarrow$ Learner 2 $\rightarrow$ Learner 3 $\rightarrow$...

Basic Design of Single-Agent Iterated Learning (Wedel 2015:153)

- ▶ See, e.g., Ito and Feldman 2022 for a recent example of language change modeled using iterated agent-based learning.

§1 Introduction: Basics of Stress Typology

§2 Empirical Basis: A Metrical Stress System with a Four-Syllable Window

- ▶ Main Case Study: Post-Vedic (Classical) Sanskrit

§3 The Learnability of Metrical Stress

- ▶ What is “hidden” structure, and how do learners cope with it?

§4 Relating Learnability and Typology: Why Should Preantepenultimate Stress Be Marked?

- ▶ What makes a four syllable window especially complex?

§5 The Role of Diachrony: Where do four-syllable windows go?

- ▶ Are four-syllable windows accurately transmitted across generations of learners?

Empirical Basis: A Metrical Stress System with a Four-Syllable Window

Four-Syllable Windows vs. Unbounded Stress

- ▶ **Important Distinction:** stress within a strict four-syllable window as opposed to UNBOUNDED stress.
- ▶ Many languages allow stress on the fourth-to-last syllable *but also farther to the left* (up to the longest prosodic word in the language).
- ▶ These are *unbounded* stress systems.
(Hayes 1995:296–99, Walker 1996, de Lacy 2002).

(6) Default-to-Same-Side (Left) Syllabic Trochees in Cupeño: (' σ σ) anywhere, or (' σ) at the right edge (Yates 2017)

- a. /nə - míx - á:n - βə/ → [nə.(¹mi.xan).βə] '(where) I did (something)'
(3SG-be-AAN-SUB.RL)
- b. /ʔə - jax - qalí/ → [ʔə.ja.qa.(¹liʔ)] 'you say'
(2SG-say-DS.SG)
- c. /pəm - təw - wəni/ → [(¹pəm.tə).wə.ni] 'we see'
(1PL-see-DS.PL)
- d. /pulif-pə-ja-qal/ → [(¹pu.liʃ).pə.ja.qal] 'he was going out'
(go.out-3SG-TH-PST.IMPV.SG)

Four-Syllable Windows vs. Unbounded Stress

(6) Default-to-Same-Side (Left) Syllabic Trochees in Cupeño: (' σ σ) anywhere, or (' σ) at the right edge (Yates 2017)

- a. /nə - míx - á:n - βə/ → [nə.(¹mi.xan).βə] '(where) I did (something)'
(3SG-be-AAN-SUB.RL)
- b. /ʔə - jax - qalí/ → [ʔə.ja.qa.(¹liʔ)] 'you say' (2SG-say-DS.SG)
- c. /pəm - təw - wəni/ → [(¹pəm.tə).wə.ni] 'we see' (1PL-see-DS.PL)
- d. /pulif-pə-ja-qal/ → [(¹pu.liʃ).pə.ja.qal] 'he was going out'
(go.out-3SG-TH-PST.IMPV.SG)

- ▶ A proper right-edge-oriented four-syllable window strictly excludes primary stress on the fifth-to-last syllable: *[' σ σ σ σ σ].
- ▶ Typologically, unbounded stress is well represented: Heinz (2009:310) reports 45 languages with quantity-sensitive unbounded stress, instantiating 26 distinct patterns.

Unbounded, Morphologically Sensitive Stress in Vedic (1)

- The earliest attested Indo-Aryan language, **Vedic Sanskrit**, also exhibits unbounded and contrastive stress.

(7) Contrastive and Unbounded Stress in Vedic

- a. Two-syllable: *úpa* ‘to(wards), by’ vs. *utá* ‘and’
- b. Four-syllable: *puróhitam* ‘domestic priest:M.ACC.SG’
vs. *purūtámam* ‘much:CMPV.M.ACC.SG’
- c. Six-syllable: *abhimātiṣāhás* ‘conquering enemies:NOM.PL’
vs. *ánipadyamānam* ‘not falling down:ACC.SG’
- d. Eight-syllable: *apratidhṛṣṭaśávasam* ‘having irresistible power:MASC.ACC.SG’
vs. *abhibhañj-at-í-nām* ‘smashing against-PART.ACT-FEM.-GEN.PL’

Unbounded, Morphologically Sensitive Stress in Vedic (2)

- Stress assignment in Vedic not only depends on lexically prespecified prosody, but also on the structure of morphologically complex stems: the HEAD (usually outermost and rightmost derivational suffix) preferentially determines the position of stress (Sandell 2023a:Ch. 4; cf. Revithiadou 1999 on Modern Greek and Russian).

(8) Stress-repelling *-tā-* vs. stress-attracting *-tvá-* and stress-attracting *-tá-*
(ACCENTUAL DOMINANCE)

- a. *bāndh-u-* + *'-tā-* $\Rightarrow$ *bandhútā-* 'kinship'
- b. *tákṣ-* + *-tá-* $\Rightarrow$ *taṣṭá-* 'fashioned' (cf. NPST.3PL *tákṣ-ati*, PST.1PL *tákṣ-ma*)
- c. *dev-á-* + *-tvá-* $\Rightarrow$ *devatvá-* 'divine power'

- The diachronically later “Classical Sanskrit stress rule” discussed below thus points to the development of a morphologically insensitive, bounded stress-assignment process out of the older Vedic word-prosodic system.

Stress in Post-Vedic Sanskrit: Evidence

- ▶ **Post-Vedic (“Classical”) Sanskrit:** convergent evidence from four sources
(see Sandell 2023a: <https://www.ryan-sandell.com/word-prosodic-systems>)
 - 1. 19th-century reports
(Haug 1874:99–100, Bühler 1883:Schrifttafel)
 - ▶ Fieldwork by German Indologists with native-speaker *paṇḍits* in Mahārāṣṭra and Gujarāt.
 - 2. Historical segmental developments (esp. syncope) in Middle and Modern Indo-Aryan languages, particularly Śauraseni Prākṛit and Gujarātī
(Jacobi 1893, Pischel 1897, Pischel 1900, Turner 1916).
 - 3. Erroneous description of accent rules of Vedic Sanskrit by at least one Indian grammarian
(*Phīṭsūtra* of Śāntanava; see Kielhorn 1866:36).
 - 4. Metrics in Epic Sanskrit, 16th c. Hindi
(Kazama 1975, Fairbanks 1992).
- 👉 At least this “Classical Sanskrit stress rule” provides solid evidence for a predictable metrical stress rule with bounded, four-syllable window.

The Classical Sanskrit Stress Rule (Sandell 2023a:177)

1. In disyllabic words, place primary stress on the initial syllable, *unless* it is light *and* the final syllable contains a long vowel or diphthong.
2. In words of three or more syllables, place primary stress on the rightmost non-final heavy syllable *up to the fourth-to-last* syllable.
3. In three- or four-syllable words, place primary stress on the initial syllable if the penult and antepenult are light.
4. In words of five syllables, place primary stress on the antepenult if the penult, antepenult, and preantepenult are all light.

Testimony from Gujarātī (1)

- ▶ **Premises:** 1) ancestor of Gujarātī possessed the “Classical Sanskrit stress rule”; 2) vowels under primary stress are never shortened or lost; 3) segmental developments should agree with position of stress as predicted by the “Classical Sanskrit stress rule”.
- ▶ See generally Turner 1916, 1921, 1962–1985 for data.
- (9) Apheresis of word-initial short vowels, unless stressed
 - a. Skt. *a. 'raṇ.yam* ‘wilderness’ > Guj. *rān* ‘forest, desert’
 - b. Skt. *'a.na.śa.nam* ‘fasting’ > Guj. *aṇsaṇ* ‘fasting’
- ▶ *Nota bene:* Reconstructed “Classical Sanskrit” stress is often contrary to attested Vedic stress, e.g., Ved. *áranya-* ‘wilderness’ $\neq$ C. Skt. *a. 'raṇ.yam*

Testimony from Gujarātī (2)

(10) [m] > [ũ] / V__V (Turner 1915)

- a. Skt. *ku.'mā.rah* 'boy' > Guj. *kũvār* 'prince'
- b. Skt. *sa.'mar.gha.kam* 'cheap' > Guj. *sõghũ*
- c. Skt. *sa.'mar.pa.ya.ti* 'delivers over' > Guj. *sõpe / sãpe* 'hand over'

(11) Otherwise, Skt. [m] > Guj. [m]

- 1. Skt. *'pañ.ca.ma.kah* 'fifth' > Guj. *pãcmo*
- 2. Skt. *'sap.ta.ma.kah* 'seventh' > Guj. *sãtmo*
- 3. Skt. *,pa.ri.'kra.ma.nam* 'walking about' > Guj. *parkamñũ*

► Ask if you would like to see more Gujarātī data!

The Classical Sanskrit Stress Rule (Sandell 2023a:177)

1. In disyllabic words, place primary stress on the initial syllable, *unless* it is light *and* the final syllable contains a long vowel or diphthong.

(12) Disyllables

- a. ('ga.ta) 'come:IMP.2PL'
- b. ('ja.la)⟨m⟩ 'water:ACC.SG'
- c. ja.('gau) 'go:PERF.1SG'

The Classical Sanskrit Stress Rule (Sandell 2023a:177)

1. In disyllabic words, place primary stress on the initial syllable, *unless* it is light *and* the final syllable contains a long vowel or diphthong.
2. In words of three or more syllables, place primary stress on the rightmost non- final heavy syllable *up to the fourth-to-last* syllable.

(13) Longer words with a stressed heavy syllable

- a. (ka.ra).('ṇē).⟨nā⟩ 'doing:INST.SG'
- b. a.('nuṣ).ṭhi.⟨tam⟩ 'performed:ACC.SG'
- c. hi.('raṇ).ya.va.⟨tī⟩ 'golden:F.NOM.SG'

The Classical Sanskrit Stress Rule (Sandell 2023a:177)

1. In disyllabic words, place primary stress on the initial syllable, *unless* it is light *and* the final syllable contains a long vowel or diphthong.
2. In words of three or more syllables, place primary stress on the rightmost non- final heavy syllable *up to the fourth-to-last* syllable.
3. In three- or four-syllable words, place primary stress on the initial syllable if the penult and antepenult are light.

(14) Three- and four-syllable words with a stressed light syllable

- a. ('bi.bhr̥).⟨ta⟩ 'bear:IMPV.2PL'
- b. ('du.hi).ta.⟨ram⟩ 'daughter:ACC.SG'

The Classical Sanskrit Stress Rule (Sandell 2023a:177)

1. In disyllabic words, place primary stress on the initial syllable, *unless* it is light *and* the final syllable contains a long vowel or diphthong.
2. In words of three or more syllables, place primary stress on the rightmost non- final heavy syllable *up to the fourth-to-last* syllable.
3. In three- or four-syllable words, place primary stress on the initial syllable if the penult and antepenult are light.
4. In words of five syllables, place primary stress on the antepenult if the penult is light.

(15) Five-syllable words

- a. (ᵢu.pa).(ˈga.ma).⟨tam⟩ ‘approach:AOR.2DU’
- b. (ᵢu.pa).(ˈgac).cha.⟨ti⟩ ‘arrive:PRS.3SG’

Exemplifying the Classical Sanskrit Stress Rule

(12) Disyllables

- a. ('ga.ta) 'come:IMP.2PL'
- b. ('ja.la)⟨m⟩ 'water:ACC.SG' > Guj. *jaḷ*
- c. ja.('gau) 'go:PERF.1SG'

(13) Longer words with a stressed heavy syllable

- a. (,ka.ra).('ṇē).⟨nā⟩ 'doing:INST.SG'
- b. a.('nuṣ).ṭhi.⟨tam⟩ 'performed:ACC.SG'
- c. hi.('raṇ).ya.va.⟨tī⟩ 'golden:F.NOM.SG'

(14) Three- and four-syllable words with a stressed light syllable

- a. ('bi.bhṛ).⟨ta⟩ 'bear:IMPV.2PL'
- b. ('du.hi).ta.⟨ram⟩ 'daughter:ACC.SG'

(15) Five-syllable words

- a. (,u.pa).('ga.ma).⟨tam⟩ 'approach:AOR.2DU'
- b. (,u.pa).('gac).cha.⟨ti⟩ 'arrive:PRS.3SG'

- Gap: No secure stress for words of six or more syllables without H in the window.

Preview: Summary of Analysis

- ▶ **Essence of the Present Analysis** (constraints after Kager 1999:Ch. 4):
 - ▶ All feet are trochaic: ('L L), ('H), maybe also ('H L) = TROCHEE.
 - ▶ Monomoraic feet are banned = FOOTBINARITY.
 - ▶ The rightmost foot in a ω receives primary stress = RIGHTMOST.
 - ▶ Feet at the right edge of a ω are avoided, mostly by not parsing final syllables into feet (NONFINALITY- $\mathcal{F}$ $\gg$ PARSE- σ).
 - ▶ In some disyllables, final consonants are not parsed into syllables in order to prevent a foot at the right edge of a ω (NONFINALITY- $\mathcal{F}$ $\gg$ PARSE-C $\gg$ PARSE- σ).
 - ▶ All heavy syllables should be stressed = WEIGHT-TO-STRESS PRINCIPLE.
 - ▶ The left edge of a ω should be aligned with the left edge of a foot ALIGN-L(ω , $\mathcal{F}$).
- 👉 The interaction between RIGHTMOST, ALIGN-L(ω , $\mathcal{F}$), and final syllable extrametricality (NONFINALITY- $\mathcal{F}$) is crucial to producing the four-syllable window.

Analyzing the Classical Sanskrit Stress Rule: /L H/ Words

(16) /jagau/ → [ja.(ˈgau)] ‘go:PERF.1SG’ (Bühler 1883)

/jagau/	TROCHEE	FTBIN	WSP	NONFIN-σ	ALIGN-L(ω, F)	ALL-Ft-L	PARSE-σ
a.  ja.(ˈgau)					*	*	*
b. (ˈja.gau)			*!				
c. (ˈja).gau		*!	*				
d. (ja.ˈgau)	*!						

- FOOTBINARITY and the WEIGHT-TO-STRESS PRINCIPLE conspire to place stress on the heavy final syllable, rather than the initial light syllable.

Analyzing the Classical Sanskrit Stress Rule: /L L/ Words

(17) /jalām/ → [(‘ja.la)⟨m⟩] ‘water:ACC.SG’ > Guj. *jaḷ* [dʒəḷ]

/jalām/		TROCHEE	FTBIN	NONFIN-f	PARSE-C	WSP	NONFIN-σ	ALIGN-L(ω, f)	ALL-FT-L	PARSE-σ
a.	जलम् (‘ja.la)⟨m⟩				*					
b.	ja.(‘lam)			*!			*	*	*	*
c.	(‘ja).lam		*!			*				*
d.	(ja.‘la)⟨m⟩	*!			*					

- The final consonant is treated as extrametrical, violating PARSE-C, to ensure both that the foot created is bimoraic and not directly aligned with the right edge of the prosodic word.

Analyzing the Classical Sanskrit Stress Rule: /L L L/ Words

(18) /ksipasi/ → [('kʂi.pa).si] 'hit:PRES.2SG (Bühler 1883)

/ksipasi/		TROCHEE	FTBIN	NONFIN- $\mathcal{F}$	PARSE-C	WSP	NONFIN- σ	ALIGN-L($\omega, \mathcal{F}$)	ALL-FT-L	PARSE- σ
a.	ꣳꣳꣳ ('kʂi.pa).si				*					*
b.	kʂi.('pa.si)			*!	*			*	*	*

- The only foot created is not at the right edge of the ω , satisfying both NONFIN- $\mathcal{F}$ and ALIGN-L($\omega, \mathcal{F}$).

Analyzing the Classical Sanskrit Stress Rule: /L L L σ / Words

(19) /duhitaram/ $\rightarrow$ [(¹du.hi).ta.ram] ‘daughter:ACC.SG’ (Bühler 1883)

/duhitaram/		RIGHTMOST	TROCHEE	NONFIN- f	PARSE-C	WSP	ALIGN-L(ω, f)	ALL-Ft-L	PARSE- σ
a.	दुहितारम् (¹ du.hi).ta.<ram>					*			**
b.	du.<(¹ hi.ta).<ram>					*	*!	*!	**
c.	<(du.hi).<(¹ ta.ra)<m>				*!		*	**	
d.	<(du.hi).ta.<(¹ ram)			*!				***	*

► ALIGN-L(ω, f) ensures that the only foot built is formed at the left edge of the ω .

Analyzing the Classical Sanskrit Stress Rule: /L L L L L/ Words

(20) /upagamatam/ → [(,u.pa).('ga.ma).tam] 'approach:AOR.2DU' (Bühler 1883)

/upagamatam/		RIGHTMOST	TROCHEE	FTBIN	WSP	ALIGN-L(ω, f)	PARSE- σ	ALL-Ft-L	ALL-Ft-R
a.	☞ (,u.pa).('ga.ma).⟨tam⟩				*		*	**	***
b.	('u.pa).(,ga.ma).⟨tam⟩	*!			*		*	**	***
c.	u.('pa.ga).ma.⟨tam⟩				*	*!	***!	*	**
d.	u.pa.('ga.ma).⟨tam⟩				*	*!	***!	**	*
e.	('u.pa).ga.ma.⟨tam⟩				*		***!		***

- Secondary stress emerges on the initial syllable, since two feet can be built, better satisfying ALIGN-L(ω, f) and PARSE- σ .

Analyzing the Classical Sanskrit Stress Rule: Words with One Heavy Syllable

(21) /hiranyavatī/ → [hi.(ˈraṇ).ya.va.tī] ‘golden:F.NOM.SG’ (*Dvirūpakoṣas*)

/hiranyavatī/		FtBIN	NONFIN-F	WSP	ALIGN-L(ω, f)	*CLASH	PARSE-σ	ALL-Ft-R	ALL-Ft-L
a.	hi.(ˈraṇ).ya.va.(tī)			*	*		****	3	1
b.	hi.(raṇ).(ˈya.va).(tī)			*	*	*!	**	4	3
c.	(hi.raṇ).(ˈya.va).(tī)			**!			*	4	2
d.	hi.(raṇ).ya.va.(tī)		*!		*		***	3	5
e.	(hi).(ˈraṇ).ya.va.(tī)	*!		*		*	***	7	1

- ▶ *CLASH blocks stress on the light antepenult as long as stress on a heavy syllable is a higher priority. Stress on the initial syllable will create violations of either WSP or FtBIN.
- ▶ A word of the shape /H L L L σ/ will be stressed [(ˌH) L (ˈL L) ⟨σ⟩] due to *CLASH and RIGHTMOST.

Analyzing the Classical Sanskrit Stress Rule: Words with Multiple Heavy Syllables

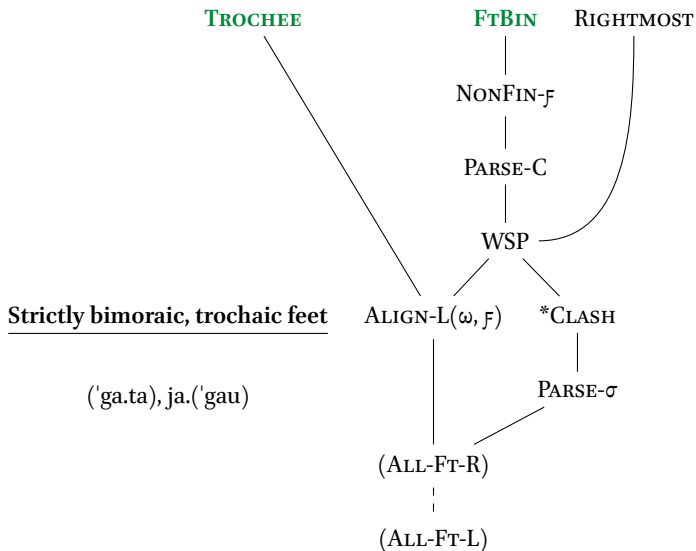
(22) /yajnopavītam/ → [(₁yaj).(₁ñō).pa.(¹vī).tam]

'investiture with sacred thread:ACC.SG' > Guj. *janōi* [dʒə.no.i]

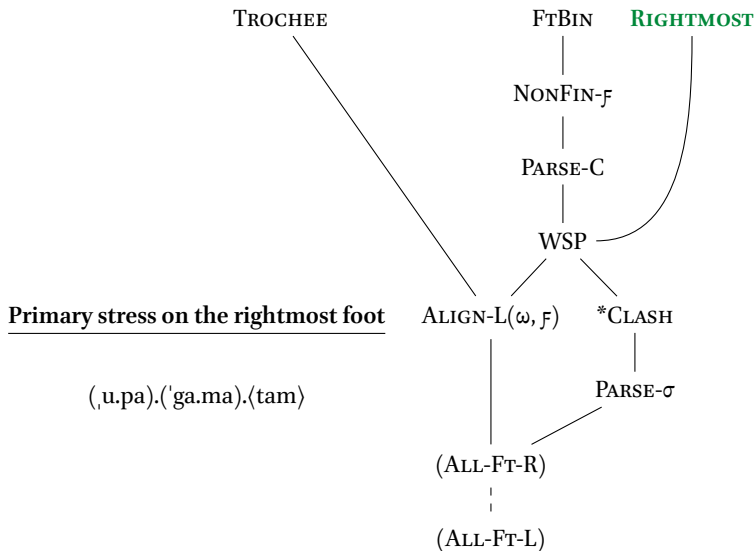
/yajnopavītam/		RIGHTMOST		ALIGN-L(ω, f)		PARSE- σ	ALL-Ft-R	ALL-Ft-L
		WSP			*CLASH			
a.	 (,yaj).(,ñō).pa.('vī).⟨tam⟩		*		*	**	7	4
b.	(,yaj).ñō.pa.('vī).⟨tam⟩		**!			***	5	3
c.	yaj.(,ñō).pa.('vī).⟨tam⟩		**!	*		***	4	4
d.	(,yaj).(,ñō).('pa.vī).⟨tam⟩		**!		**	*	7	3
e.	('yaj).(,ñō).pa.('vī).⟨tam⟩	*!	*		*	**	7	4

- All heavy syllables receive stress, even in clash, and the rightmost foot is selected for primary stress.

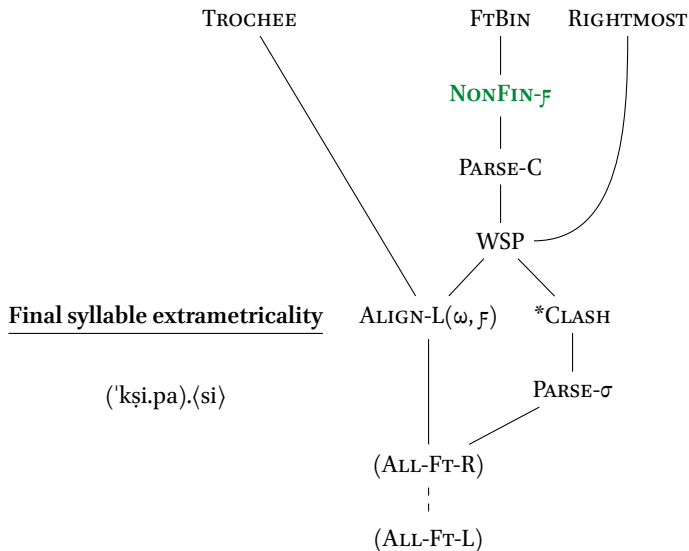
Hasse Diagram of Final Total Constraint Ranking: Takeaways



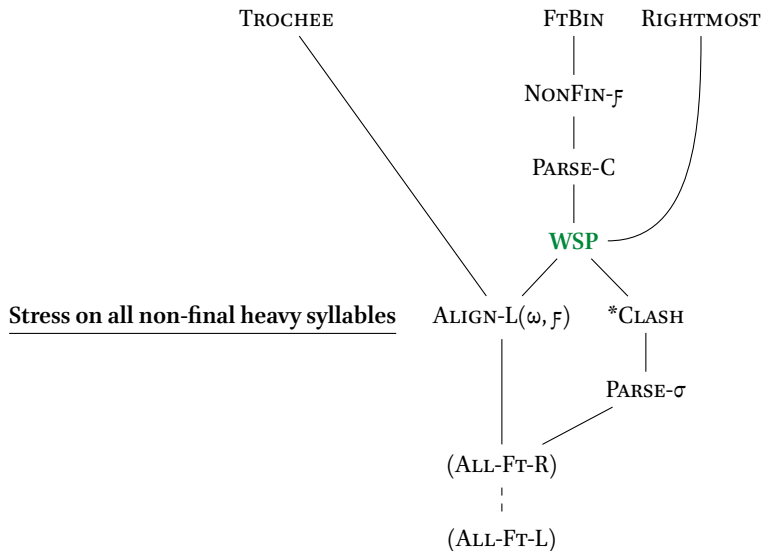
Hasse Diagram of Final Total Constraint Ranking: Takeaways



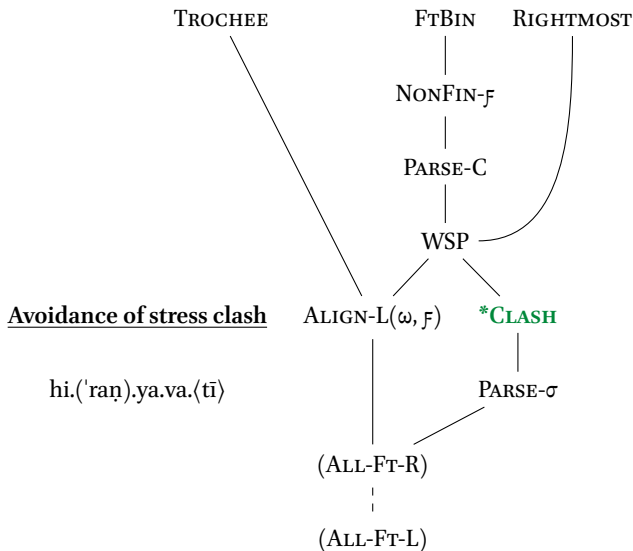
Hasse Diagram of Final Total Constraint Ranking: Takeaways



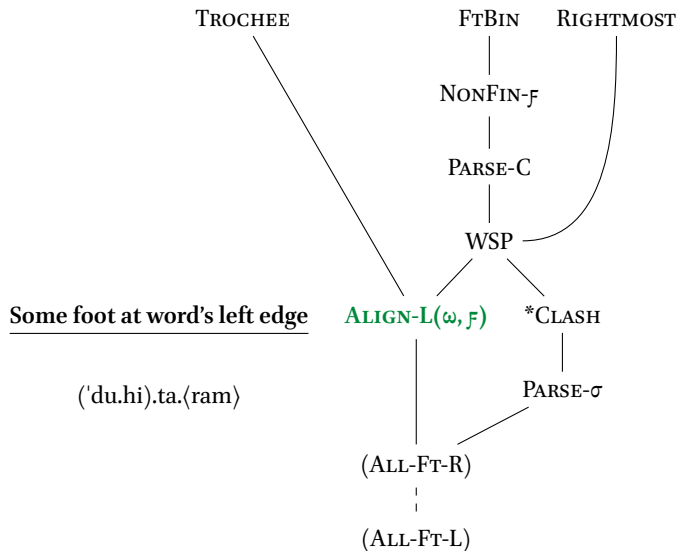
Hasse Diagram of Final Total Constraint Ranking: Takeaways



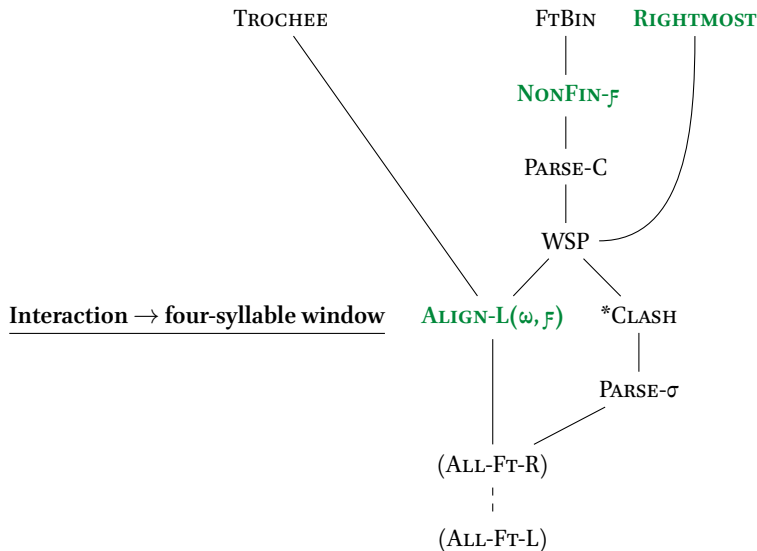
Hasse Diagram of Final Total Constraint Ranking: Takeaways



Hasse Diagram of Final Total Constraint Ranking: Takeaways



Hasse Diagram of Final Total Constraint Ranking: Takeaways



Outstanding Questions

- ▶ If stress patterns like that of the “Classical Sanskrit stress rule” apparently *can* exist, why do they come about so rarely?
- ▶ To begin to answer this question, we need to consider how grammars for stress assignment can be learned from (largely) unstructured data.

The Learnability of Metrical Stress

What Does “Learnable” Mean?

- ▶ Central questions for learnability (cf. Heinz 2009:307):
 1. How is the hypothesis space structured?
 2. How does a learner go about searching the hypothesis space, making inferences from (positive) data?
- ▶ Clarifying 1. yields insights on what is typologically possible at all and how languages (formally) differ.
- ▶ Clarifying 2. helps to understand how languages move within the typology.
 - ⇒ Why are certain grammars are more common?
 - ⇒ What sorts of diachronic changes are possible?

The Learner's Perspective: Searching the Hypothesis Space

- ▶ **Objective:** Accurately reproduce the primary linguistic data to which the learner is exposed.
 - ▶ Caution: Even if the grammar $\mathcal{G}$ to be acquired is learnable, the way in which the learner searches the hypothesis space might not be guaranteed to arrive to the same grammatical analysis as the grammar $\mathcal{G}$ that generated the data.
- ▶ **Two Questions** (for learning metrical stress):
 1. How are constraints that select the optimal candidate (or govern the probability of the candidates) to be ranked or numerically weighted?
 2. How do I decide what structures are present in the data when they are not transparent on the surface?
 - ▶ This problem is acute for prosodic structures like FEET, which themselves have no direct acoustic correlates.

Learning Constraint Weights: Maximum Entropy Grammar (1)

- ▶ Maximum Entropy modeling allows for grammars in which the same input has more than one grammatically licit output (i.e., variation).
 - ▶ Mathematically, MaxEnt is equivalent to multinomial logistic regression.
- ▶ Hayes and Wilson (2008:386): “maximizing entropy is in fact equivalent to maximizing the probability of the observed forms given the constraints.”
- ▶ MaxEnt grammars thus generate probability distributions over a set of candidates, given some underlying representation: each candidate has some probability of winning.
 - ▶ Ask if you would like to see an example of how candidate probabilities are calculated.
- ▶ The target pattern to be learned (“C. Skt. stress rule”) is categorical, but learning with MaxEnt because:
 1. Learning is not “all or nothing”: a pattern might be partly learned, but not completely, i.e. imperfectly.
 2. We can diagnose which specific aspects of the pattern pose challenges for a learner (instead of simply seeing a global failure to learn the grammar).

Learning Constraint Weights: Maximum Entropy Grammar (3)

- ▶ But how are the constraint weights learned?
- ▶ One approach: STOCHASTIC GRADIENT ASCENT (online learning; Jäger 2007).

$$w_{new} = w_{old} + \Delta w_i, \text{ where } \Delta w_i = \eta \times (L_i - W_i)$$

This is the Perceptron update rule: Rosenblatt 1958

- ▶ When the learner's predicted winner L_i does not match the winner indicated by the learning datum, W_i , an update in constraint weights in favor of W_i will take place.
- ▶ **Effect:** Expected winners increase in probability with increased exposure to data.
- ▶ Ask me later if you are curious about details of the learning algorithm.

The Hidden Structure Problem

► Two issues in learning the position of primary stress:

1. Relatively shorter words are not very informative, because they are compatible with multiple different grammatical hypotheses (at least individually).

⇒ Is a string $['\sigma \sigma \sigma]$ indicative of initial stress, antepenultimate stress, or maybe something more complex?

2. Given that stress is dependent upon foot structure, what foot structure(s) might be responsible for the surface stress (Dresher and Kaye 1990)?

⇒ Given a string $[L 'H L]$, is the right parse $[(L 'H) L]$, $[L ('H) L]$, or $[L ('H L)]$?

► 1. can become a serious issue if crucially disambiguating data are accidentally rare.

► 2. is a *hidden structure* problem (Tesar and Smolensky 1998, Tesar and Smolensky 2000, Jarosz 2013, Jarosz 2016, Jarosz 2019:75–81).

- **Credit Problem:** When I hear $[L 'H L]$, which constraints deserve credit for the fact that stress is on the medial heavy syllable (Dresher 1999:54–8)?

ROBUST INTERPRETIVE PARSING (1)

- ▶ ROBUST INTERPRETIVE PARSING (RIP; Tesar and Smolensky 1998) tries to escape from the Credit Problem by allowing the learner to make an “educated guess” about what the correct parse of a learning datum is likely to be, given what the learner already thinks is true about their grammar.
- 👉 ROBUST INTERPRETIVE PARSING is demonstrably effective at overcoming hidden structure problems in a variety of constraint formalisms = progress! (Tesar and Smolensky 2000, Jarosz 2013, Jarosz 2016, Boersma and Pater 2016, Nazarov and Jarosz 2017)
- ▶ Ask me for an example of how RIP functions in practice.

**Relating Learnability and Typology:
Why should preantepenultimate stress be marked?**

- ▶ Use computational simulation to assess *whether* a stress pattern is learnable at all, and if so, how *quickly* and *easily* different classes of stress patterns may be learned.
 - ▶ Compare Tesar and Smolensky 2000:Ch. 4, Staubs 2014, Stanton 2016, Boersma and Pater 2016, Nazarov and Jarosz 2017.
- ▶ **Claims:**
 1. The Classical Sanskrit stress pattern is *learnable*: learning agents trained on overt surface forms (like ['L L L L], [L L L 'H H], etc.) *can* learn a weighted constraint grammar that reproduces the pattern categorically.
 2. However: The Classical Sanskrit stress pattern is demonstrably more complex and challenging to learn than a variety of other metrical stress patterns.

The Classical Sanskrit Stress Rule is Learnable

- ▶ **Learnability Simulation:** Robust Interpretive Parsing (Tesar and Smolensky 1998) under MaxEnt Stochastic Gradient Ascent (Jäger 2007).
 - ▶ Unification of tools in phonological analysis for simultaneously learning grammars with **variation** and **hidden structure**.
- ▶ In this simulation, prosodic shapes were restricted to those 2–5 syllables in length that actually occur in a corpus of Vedic Sanskrit (*Ṛgveda*).
 - ▶ All logically possible combinations of 2–4 syllables are attested; some gaps occur among the 5-syllable words.

Learnability Simulation: Design

- ▶ Simulation coded and run in R
(Version 4.4.2; R Core Development Team 2024)
- ▶ Constraints: identical to the constraints used for the categorical grammar in Section 2.
 - ▶ All constraint weights initially set to 5 and given a floor of 0 (cf. Hayes 2004).
- ▶ Winner Selection: Maximum Entropy
- ▶ Update Rule for Constraint Weights: Symmetric Perceptron with $\eta = 0.1$
(Rosenblatt 1958)
- ▶ Training Data: between 1000 and 100000 tokens per run (100 runs at each N)
 - ▶ 2–5 syllable word forms marked for primary stress distinguishing Heavy and Light syllables, *as well as* four different types of final syllables (V, VC, V:, and V:C).
 - ▶ All word shapes are equally probable in the training data (avoid accidental exclusion of prosodic shapes) and are sampled randomly.

Tableaux for Learnability Simulation

		Freq	Parse-Seg	Parse-Syllable	FtBin	NonFinality	NonFinality-Foot	NonInitiality-Ft	WSP	Trochee	lamb	All-Ft-L	All-Ft-R	Align-wd-L	Align-wd-R	Rh-Contour
		Freq	Parse-Seg	Parse-Syllable	FtBin	NonFinality	NonFinality-Foot	NonInitiality-Ft	WSP	Trochee	lamb	All-Ft-L	All-Ft-R	Align-wd-L	Align-wd-R	Rh-Contour
/L V/	[L1 V] \-> (L1 V)		1				1	1				1				
	[L V1] \-> (L V1)					1		1								
	[L1 V] \-> (L1) V				1 1			1			1			1		
/L V:/	[L V1] \-> L (V1)				1 1	1		1					1		1	
	[L1 V:] \-> (L1 V:)							1	1			1				
	[L V:] \-> (L V:)					1		1			1					
/L V:C/	[L1 V:] \-> (L1 V:)				1 1			1	1					1		
	[L V:] \-> L (V:)	1			1	1		1					1		1	
	[L1 VC] \-> (L1 VC)						1	1	1							
	[L1 VC] \-> (L1) V:C	1	1					1				1				
	[L V1C] \-> (L V1C)					1		1			1				1	
	[L V1C] \-> (L V1)C			1			1	1			1					
	[L1 VC] \-> (L1) VC				1 1			1	1					1		
	[L V1C] \-> L (V1)C				1	1		1					1		1	
	[L V1C] \-> L (V1)C			1	1	1		1				1			1	
/L V:C/	[L1 V:C] \-> (L1 V:C)						1		1	1						
	[L1 V:C] \-> (L1 V:):C			1				1	1			1				
	[L V:]C \-> L (V:):C						1	1			1					
	[L V:]C \-> L (V:):C						1	1			1					
	[L V:]C \-> (L1) V:C				1 1			1	1				1		1	
	[L V:]C \-> L (V:):C				1	1		1				1			1	
	[L V:]C \-> L (V:):C				1	1		1				1			1	
	[L V:]C \-> L (V:):C	1	1		1	1		1				1			1	

- ▶ 118 tableaux with 1736 distinct candidates.
- ▶ Candidates and constraint violations generated through scripting in R.

Learnability Simulation: Quantitative Assessment 1

- ▶ **Sum squared error (SSE)** between the probabilities of candidates under a categorical Classical Sanskrit stress grammar and the probabilities assigned by the weights at the end of the simulation.
 - ▶ Mean SSE over 100 runs trained on 80000 tokens each: **4.79**
⇒ approaching categoricity, but not consistently by the termination of training.
 - ▶ In 19/100 simulations at this size, SSE was less than 1 (= virtual categoricity).
 - 👉 These are “successes” ⇒ **indicative of learnable pattern.**

Learnability Simulation: Quantitative Assessment 2

- Mean SSE across 100 simulations with training data in any amount is appreciably higher than the mean SSE for range of other predictable primary stress patterns tested.
- ⇒ Learning of the Classical Sanskrit pattern is overall more challenging.

N	QI Initial	QI Final	QI Penult	Right Trochees	C. Lat	C. Skt.
1000	0.13	0.18	0.79	1.21	3.37	12.74
2000	0.03	0.09	0.33	0.45	0.89	9.97
5000	0.00	0.03	0.08	0.12	0.20	8.38
10000	0.00	0.01	0.02	0.04	0.07	7.93
20000	0.00	0.00	0.00	0.02	0.02	7.42
30000	0.00	0.00	0.00	0.01	0.01	7.38
40000	0.00	0.00	0.00	0.01	0.01	7.05
50000	0.00	0.00	0.00	0.01	0.01	6.72
60000	0.00	0.00	0.00	0.00	0.00	5.59
70000	0.00	0.00	0.00	0.00	0.00	5.29
80000	0.00	0.00	0.00	0.00	0.00	4.79
90000	0.00	0.00	0.00	0.00	0.00	4.64
100000	0.00	0.00	0.00	0.00	0.00	4.17

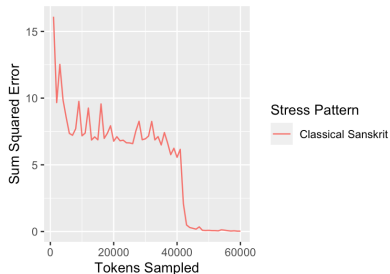
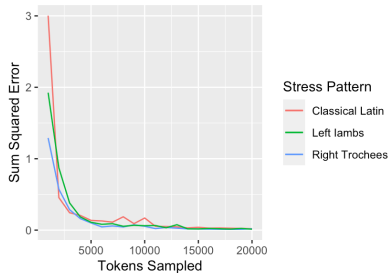
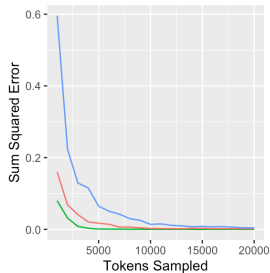
Learnability Simulation: Qualitative Assessment

- ▶ Which (if any) inputs have probabilities that diverge substantially from the expected winners under the categorical grammar ($N = 80000$, 100 runs)?
 - ▶ Initial and final stress in disyllables is learned accurately (high weight learned for FTBIN) $> 98\%$ in all simulations.
 - ▶ Stress in words with at least one heavy non-final syllable is learned accurately (high weight for WSP) $> 98\%$ in all simulations.
 - ▶ Antepenultimate stress in 3- and 5-syllable words ($/L L \sigma/$ and $/L L L L \sigma/$) is learned accurately (NONFINALITY- $f \gg$ ALL-FT-R) $> 98\%$ in all simulations.
- ▶ **Problem:** preantepenultimate stress in 4-syllable words of the shape $/L L L \sigma/$ (e.g., $[('du.hi).ta.(ram)]$)
 - ▶ In the worst performing run, $[L L L \sigma]$ receives only around 10% probability, but in the best case, it receives over 99%.
 - ▶  The extent to which $[L L L \sigma]$ can be accurately generated depends on receiving enough of the right kind of data!

Simulation 1: Quantitative Assessment 3

- ▶ How many tokens of learning data needed before a plateau in learning is reached?
 - ▶ Quantity-insensitive patterns with smaller windows need less data before achieving categoricity
(replicates and confirms Staubs 2014 using different model parameters).
 - ▶ For these patterns, they are essentially at ceiling ($SSE < 0.01$) by $N = 10000$.
- ▶ Applies to quantity-sensitive languages tested here, too (new result!): greater permissible distance from a word edge $\Rightarrow$ more learning data needed.
 - ▶ Except for the C. Skt. pattern, these are at ceiling ($SSE < 0.01$) by $N = 50000$.

Learnability Simulation: Results



Learnability Simulation: Global Summary

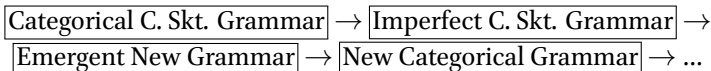
1. Simulation using a form of Robust Interpretative Parsing with MaxEnt grammars allows computational learning agents to overcome the hidden structure problem and acquire a categorical grammar that corresponds to the Classical Sanskrit pattern.
2. Learning constraint weights that yield the expected Classical Sanskrit pattern is **challenging**: sometimes learning is successful, sometimes not.
 - ▶ This finding contrasts with findings for other predictable metrical stress patterns that were tested, which are nearly always categorically learned.
3. In order for learning of a categorical Classical Sanskrit stress pattern to succeed, relatively **more** learning data is required.
 - ▶ In order to guarantee categorical learning of the Classical Sanskrit pattern, data of a larger order of magnitude is required.

**The Role of Diachrony:
Where do four-syllable windows go?**

Diachronic (In)Stability?

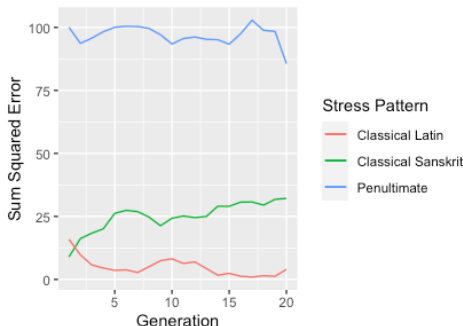
- ▶ The results from the simulation above already suggest that, in comparison to other stress patterns that, even given an original categorical pattern of data matching the Classical Sanskrit stress pattern, getting it right is challenging.
 - 👉 These data are particularly susceptible to *reanalysis* as instantiating some other stress pattern!
- ▶ **Iterative agent-based simulation:** design similar to the learnability simulation above, but:
 - ▶ The probability of candidates at the end of a generation $\mathcal{G}_i$ becomes the probability of a given overt form for a given input in the next subsequent generation $\mathcal{G}_{i+1}$.
 - ▶ E.g., if /L L L L/ is realized as [('L L) L L] with 90%, [L ('L L) L] with 10% by $\mathcal{G}_i$, then the learner in $\mathcal{G}_{i+1}$ is expected to encounter ['L L L L] and [L 'L L L] with corresponding probabilities.
 - ▶ Data is transmitted iteratively for some prespecified number of generations.

Iterated Learning and Variable Word Prosody



- ▶ **Question:** Do *non-categorical* stress grammars with some kind of variation exist / can they be acquired at all?
- ▶ Both interspeaker and intraspeaker synchronic variation in word-level prosodic features has been reported:
 - ▶ Sondregger and Niyogi 2013 on English stress
 - ▶ Lee, Jongman, and Zhang 2016 on tone South Kyengsang Korean
 - ▶ Cairene Radio Arabic with free variation in position of primary stress, reported by Harrell (1960) (cf. Hayes 1995:130–132)
- 👉 Variation as a transitional state between categorical grammars is a reasonable expectation, even where stress is concerned.

Diachronic Window Shrinkage



👉 This might mirror the history of Latin!

- ▶ Old Latin with four-syllable window $[' L L L \sigma] \gg$ three-syllable window in Classical Latin.

👉 Diachronic window shrinkage probably attested in Indic, too!

- ▶ Right-edge, quantity-sensitive three-syllable window in Standard Formal Hindi (Pandey 2021), fixed initial stress in Gujarātī (Bowers 2019).

Whence the Classical Sanskrit Stress Rule?

- ▶ Could knowledge of the history behind the Classical Sanskrit stress rule shed light on its origins?
 - ▶ If this stress grammar is difficult to acquire, perhaps some specific properties of its ancestor facilitated its emergence?
- ▶ Great luck: we know a *lot* about the word prosody of **Vedic Sanskrit** (approximate ancestor of attested Classical Sanskrit and Middle Indic languages)!
 - ▶ *prātiśākhya*s (manuals of phonetics to accompany the Vedas; see Whitney 1869)
 - ▶ Indian Grammatical tradition: Pāṇini (von Böhtlingk 1845)
 - ▶ Indologist / Indo-Europeanist descriptions: Wackernagel 1896:284–95, Macdonell 1910:76–107, Kurylowicz 1968:7–21, Jasanoff 2017:7–10
 - ▶ Formal analyses: Kiparsky 2010, Sandell 2015:Ch. 7, Perry and Vaux 2018 Yates 2020, Sandell 2023a:Ch. 4

What Happened to the Classical Sanskrit Stress Rule?

- ▶ Hindi: different reports (perhaps different dialects / registers; cf. Ohala 1977, Pandey 2021):
 - ▶ some might essentially preserve the Classical Sanskrit pattern (Grierson 1895, Fairbanks 1992).
 - ▶ some may have an unbounded, default-to-opposite pattern (Kelkar 1968).
 - ▶ three-syllable stress window, comparable to Classical Latin, but with weight scale CV:C ≫ CV:, CVC ≫ CV (Pandey 2021).
- ▶ Gujarātī: also different reports, but recent instrumental evidence points to fixed initial stress (Shih 2018, Bowers 2019).
- ▶ *Nota bene*: Not all present-day Indic languages are descendants of the variety of Old Indic (~ Sanskrit) that developed the Classical Sanskrit stress rule. The relevant varieties were probably (ancestors of) Central Middle Indic (Śauraseni Prākṛit; see Southworth 1958, Misra 1967, Ollett 2012). Others may have maintained Vedic-like lexical + morphological stress into Middle Indic (Pischel 1897, 1900).

Is it possible to bridge the gap between Vedic Sanskrit and Classical Sanskrit?



CLASSICAL SANSKRIT STRESS RULE



Is it possible to bridge the gap between Classical Sanskrit and Modern Indic languages?

Continuing Research: Vedic Sanskrit

- ▶ Recall: Vedic Sanskrit exhibits unbounded, morphologically sensitive lexical stress.
- ▶ A grammar + lexicon without lexical stress (like the C. Skt. rule) must merely learn the correct ranking of the relevant MARKEDNESS constraints.
- ▶ A grammar + lexicon *with* lexical stress requires:
 - a. special prosodic marking in the lexicon (i.e., which morphemes should receive stress or not).
 - b. active faithfulness constraints (e.g., MAX-PROMINENCE) that ensure that lexical prosodic preferences are (at least sometimes) preserved in outputs.
- ▶ Challenges for the unsupervised learning of Vedic-like stress systems from unlabeled data:
 - a. Segmentation of morphologically complex words (Baayen and Schreuder 2000, Hammarström and Borin 2010, Bergmanis and Goldwater 2017).
 - b. Identification of HEADS in morphologically complex stems.
 - c. Learning of underlying representations with lexical accent (Tesar 2004, Apoussidou 2007, Tesar 2014).

Continuing Research: Modern Indo-Aryan Languages

- ▶ Modern Indo-Aryan (ModIA) languages present both (a) considerable diversity in word prosody and (b) a series of diachronic natural experiments in the acquisition of word prosody.
 - ▶ Fixed metrical stress patterns are predominant (see, e.g., descriptions in Cardona and Jain 2003)
 - ▶ But contrastive lexical stress patterns occur, too (e.g., two-syllable left-edge window in Assamese; Goswami and Tamuli 2003:409).
 - ▶ And lexically contrastive tone in several languages, which exhibit tone $\Leftrightarrow$ stress interactions (e.g., Punjabi, Palula; Dhillon 2010).
- ▶ The effects of various factors on the reanalysis and emergence of stress systems can be tested with an iterated agent-based modeling paradigm:
 - ▶ Segmental sound changes (e.g., syncope and apocope in Gujarātī) — how do learners reanalyze stress when syllables are lost (cf. Bowers 2015 on Odawa)?
 - ▶ Tonogenesis (e.g., in Punjabi) — how do learners accommodate preferences for aligning stress and tone (cf. de Lacy 2002)?
 - ▶ Language contact — how do learners react to incompatible variation (cf. Sandell 2023a:Ch. 7 on Germanic)?

Summary and Conclusions

In this talk, we have explored the following questions:

1. Do there exist metrical stress systems that assign primary stress within a four-syllable window?
2. Why might systems with a four-syllable window be (extremely) rare?
3. Empirically, how did stress assignment work in (a variety of) post-Vedic Sanskrit?
4. What approaches to the learnability of metrical stress are available?
5. Can existing proposals for constraint inventories and approaches to learnability cope with four-syllable windows?
6. How does the learnability of a four-syllable window relate to prosodic typology and prosodic change?

Conclusions

- ▶ **Facts and Theory:** Post-Vedic (“Classical”) Sanskrit indeed possessed a four-syllable stress window, and its analysis is compatible with standard constraints on metrical stress.
- ▶ **Typology:** four-syllable windows of any kind are rare, and four-syllable windows that are due to foot structure alone seem to be exceedingly uncommon.
- ▶ **Learnability:** although learnable with sufficient data, systems with four-syllable windows are both more complex and more difficult to learn than systems with smaller windows.
- ▶ **Diachrony:** iterated agent-based learning indicates that the Classical Sanskrit stress rule is liable to become a “simpler” stress pattern.

- ▶ Tying Everything Together: learnability shapes diachrony, which yields typology.
 - ▶ In the case of four-syllable stress windows, a type of ANALYTIC BIAS (Moreton 2008) — more specifically, simplicity bias – makes them rare.
 - ▶ Especially if crucial data from 4- and 5-syllable words is encountered *too infrequently*, reanalysis as a pattern with a three-syllable window is likely.
 - ▶ **Continuing Research:**
 - ▶ Does unbounded lexical stress with a left-edge frequency bias (as in Vedic Sanskrit) present a plausible diachronic pathway to a four-syllable window?
 - ▶ How do we relate the word prosodic systems of modern Indic languages to the Classical Sanskrit stress rule / Vedic lexical stress?

Thank you!

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1. Testimony of Gujarātī / Examples of stress-induced syncope from Sandell 2023b
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The Testimony of Gujarātī

- ▶ Four-syllable [${}^{\prime}\sigma$ L L σ]
 - a. Skt. **śva.śu.ra.kaḥ* ‘father-in-law’ > Guj. *sasro*
 - b. Skt. **gar.bhi.ṇi.kā* ‘pregnant woman’ > Guj. *gābhṇi*
- ▶ Five-syllable [${}_{\text{L}}$ L ${}^{\prime}$ L L σ] (unstressed [i] > [a])
 - a. Skt. *pa.ri.ṇa.ya.ti* ‘leads around’ > Guj. *paraṇṇē* ‘marry’
 - b. Sk. *pri.ya.kā.ra.kaḥ* ‘congenial’ > Guj. *piāro*

Plausible Stress-Induced Syncope

Table 2. Plausible examples of stress-induced syncope in Sanskrit, Śaurasenī, and Gujarātī

Prosodic shape	Vedic form	Stressed Sanskrit preform	Attested form	Gloss
[L H L L σ]	<i>hiraṇyavatī-</i> (RV)	<i>hi. 'raṇ.ya.va.tī-</i>	Skt. <i>hiraṇvatī-</i> (MBh. 5×)	'possessing gold (river name)'
[L H L L σ]	<i>hiraṇyamāya-</i> (ŚB 13.2.10.2)	<i>hi. 'raṇ.ya.ma.ya-</i>	Skt. <i>hiraṇmāya-</i> (TS 5.2.7.2)	'made of gold, golden'
[L L σ]	<i>bhagīnī-*</i>	<i>'bha.gi.nī-</i>	Skt. <i>bhagnī</i> (Dvirūp.)	'sister; woman'
[H L L σ]	—	<i>'pū.ga.pha.la-</i>	Ś <i>popphalī</i>	'betel-nut'
[L H L σ]	<i>pácāmahe</i>	<i>pa. 'cā.ma.he</i>	Ś. <i>pacamha</i>	'cook:PRS.MID.1PL' > 'cook:IPV.1PL'
[L VC]	<i>jalām</i>	<i>'ja.lam</i>	G. <i>'jaḷ</i>	'water'
[L H σ]	<i>āraṇyam</i>	<i>a. 'raṇ.yam</i>	G. <i>'rāṇ</i>	'wilderness'
[L L L σ]	—	<i>'ku.ma.la.kam</i>	G. <i>kumḷo</i>	'lotus fiber' (?) ^a
[L H L L σ]	<i>cāturguṇakaḥ*</i> (cf. <i>cāturguṇa-</i> , ŚB 3.3.2.9)	<i>ca. 'tur.gu.ṇa.kah</i>	G. <i>'cōgṇo</i>	'four(fold)'
[L L L L σ]	<i>pāri ṇayati</i>	<i>,pa.ri. 'ṇa.ya.ti</i>	G. <i>'parne</i> ^b	'lead around:CAUS.3SG = marry'

Maximum Entropy Grammar: Example

- (23) Each candidate is a vector of constraint violations.

INPUT			$\alpha (w = 3)$	$\beta (w = 2)$	$\gamma (w = 1)$
a.	 -3	A	1×3		
b.	-4	B		2×2	
c.	-4	C		1×2	2×1
d.	-4	D	1×3		1×1

- ▶ Maximum Entropy (MaxEnt) grammars generate probability distributions over a set of candidates.
 - ▶ The disharmony $\mathcal{H}$ of a candidate x , is the sum of the products of the weight of a constraint, w_i by the number of violations of that constraint by the candidate x , $f_i(x)$.
 - ▶ Candidate A has a probability of ~ 0.475 given its disharmony score of 3, while all other candidates have probabilities of ~ 0.175 .
 - ▶ These probabilities are obtained by taking the exponential of a candidate's disharmony, normalized by summing the exponential of the disharmony scores for all candidates.

$$\frac{\exp(-3)}{\exp(-3) + \exp(-4) + \exp(-4) + \exp(-4)}$$

Learning Constraint Weights with Stochastic Gradient Ascent

- ▶ But how are the constraint weights learned?
- ▶ One approach: STOCHASTIC GRADIENT ASCENT (online learning; Jäger 2007).
 1. Sample an output, $o(\textit{teacher})$, from a set of training data, according to the empirical probabilities of the tokens.
 2. Find the input i associated with $o(\textit{teacher})$.
 3. Use the learner's current grammar to generate the probability of all possible outputs for the input i , and sample one of those outputs $\rightarrow o(\textit{learner})$.
 4. If the learner's "winner", $o(\textit{learner})$ is not the same as the token of data, $o(\textit{teacher})$, then make a small adjustment in constraint weights, w :
$$w_{\textit{new}} = w_{\textit{old}} + \Delta w_i, \text{ where } \Delta w_i = \eta \times (L_i - W_i)$$

This is the Perceptron update rule: Rosenblatt 1958
- ▶ The weights of constraints violated by $o(\textit{learner})$ but not by $o(\textit{teacher})$ become higher, and the weights of constraints violated by $o(\textit{teacher})$ but not by $o(\textit{learner})$ become lower.

Robust Interpretive Parsing for Metrical Stress

- ▶ ROBUST INTERPRETIVE PARSING (RIP; Tesar and Smolensky 1998) tries to escape from the Credit Problem by allowing the learner to make an “educated guess” about what the correct parse of a learning datum is likely to be, given what the learner already thinks is true about their grammar.
 - ▶ For metrical stress, since the position of stress is due to markedness alone, the UR for a word is a equivalent to its (syllabic) string, minus stress (e.g., $[\sigma '\sigma \sigma] = |\sigma \sigma \sigma|$)
 - ▶ The learner can generate all logically possible parses, which are candidates (e.g., for $|\sigma \sigma \sigma|$: $/(' \sigma \sigma) \sigma /$, $/(\sigma ' \sigma) \sigma /$, etc.).
 - ▶ Given a specific learning datum, e.g., $[\sigma '\sigma \sigma]$, the set of logically possible parses compatible with the overt information constitute the robust parses = for $[\sigma '\sigma \sigma]$: $/(\sigma ' \sigma) \sigma /$, $/\sigma (' \sigma \sigma) /$, $/\sigma (' \sigma) \sigma /$.

ROBUST INTERPRETIVE PARSING: Example

(20) RIP applied to the learning datum $[\sigma' \sigma \sigma]$

(cf. Jarosz 2013:35–6, Jarosz 2016:207–9)

	$ \sigma\sigma\sigma $	ALL-FT-L	TROCHAIC	IAMBIC	ALL-FT-R
a.	 $[\sigma'\sigma\sigma] \Rightarrow /(\sigma'\sigma)\sigma/$			*	*
b.	RIP $[\sigma'\sigma\sigma] \Rightarrow /(\sigma'\sigma)\sigma/$		*!		*
c.	$[\sigma'\sigma\sigma] \Rightarrow / \sigma(\sigma'\sigma)/$	*!		*	
d.	$[\sigma\sigma'\sigma] \Rightarrow / \sigma(\sigma'\sigma)/$	*!	*		

(21) RIP applied to $[\sigma' \sigma \sigma]$ after constraint demotion

$\Rightarrow$ RIP = learner's optimal parse

	$ \sigma\sigma\sigma $	ALL-FT-L	IAMBIC	ALL-FT-R	TROCHAIC
a.	$[\sigma'\sigma\sigma] \Rightarrow /(\sigma'\sigma)\sigma/$		*!	*	
b.	 , RIP $[\sigma'\sigma\sigma] \Rightarrow /(\sigma'\sigma)\sigma/$			*	*
c.	$[\sigma'\sigma\sigma] \Rightarrow / \sigma(\sigma'\sigma)/$	*!	*		
d.	$[\sigma\sigma'\sigma] \Rightarrow / \sigma(\sigma'\sigma)/$	*!			*

Emergence of Final-Syllable Extrametricality?

- Recall that the C. Skt. stress pattern disallows final stress, except in disyllables of the shape [L V:(C)].
- Final stress in Vedic Sanskrit happens to be significantly underrepresented:

	2-syllable	3-syllable	4-syllable	5-syllable
Expected % (proportional distribution)	0.5	0.3	0.25	0.2
Empirical % Final Stress	48%	20.7%	12.8%	4.1%
p -value (1-sample proportions test)	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$
χ^2 -value	81.5	2775	1128.3	463.11

Empirical vs. Unbiased Distribution of Final Stress in the *R̥gveda* (by Token)

Overrepresentation of Stress on Heavy Syllables in Vedic

(22) Distribution of Stress Position in Disyllables by Weight of Initial Syllable and Length of Final Vowel (Type Frequencies)

a. Stress Position by Weight of Initial Syllable

	Initial H	Initial L	Total	
Initial Stress	1923	677	2600	
Final Stress	1940	830	2770	$\chi^2 = 10.043$
Total	3863	1507	5370	$p = 0.0015$

- Stress is somewhat overrepresented on initial heavy syllables and underrepresented on initial light syllables.
- Effect size is very weak (Cramér's $V = 0.043$).

b. Stress Position by Vowel Length in Final Syllable

	Final V:(C)	Final V(C)	Total	
Initial Stress	1127	1473	2600	
Final Stress	1482	1288	2770	$\chi^2 = 54.968$
Total	2602	2761	5370	$p < 0.001$

- Stress is overrepresented on final syllables with a long vowel or diphthong.
- Effect size is small (Cramér's $V = 0.1$).